

SUMMARY

Robotics and control systems engineer with hands-on experience building and field-testing ROS2-based autonomous systems. Architected full autonomy stacks from perception through actuation—integrating LiDAR, stereo cameras, IMU, and encoders into real-time closed-loop control pipelines at sub-10 ms latency. Experienced in multi-loop control architecture (position/velocity/torque), sensor fusion, and structured systems design. Published researcher (IFTToMM, IEEE RO-MAN, IC2E3) and international competition winner.

TECHNICAL SKILLS

|                 |                                                                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------------------|
| Robotics Stack  | ROS2 (Nav2, TF2, SLAM, sensor drivers), Gazebo, RViz, custom node architecture, topic/service/action interfaces         |
| Languages       | C++ (primary — real-time control), Python (perception, tooling, data analysis), MATLAB/Simulink                         |
| Control Systems | Multi-loop PID (position/velocity/torque), impedance control, adaptive control, gain scheduling — all on real hardware  |
| Perception      | Stereo vision (ZED 2i, D435i), Livox LiDAR, visual odometry, CLAHE, YOLO-based detection, IMU sensor fusion             |
| Hardware        | LiDAR, depth cameras, IMU, encoders, EMG sensors, force plates; electrical schematics, sensor wiring, hardware bring-up |
| Dev & Tooling   | Linux (Ubuntu), Git, distributed IPC, latency profiling & optimisation                                                  |

EXPERIENCE

Controls & Robotics Engineering Intern  
CoDSAI — UQ-TIET Joint Lab

Jun 2025 — Present  
Patiala, Punjab

- Designed and deployed a 100 Hz real-time multi-loop control system (inner torque / outer position loop) for a lower-limb exoskeleton operating in human-interactive environments.
- Integrated heterogeneous sensor suite—EMG, AMTI force plates, Qualisys motion capture—into a unified closed-loop pipeline; implemented intent-estimation logic for adaptive assistive motion.
- Reduced joint tracking error from 10° to <5° through impedance controller tuning and adaptive gain scheduling; validated against performance specifications prior to deployment.
- Authored control architecture specifications and commissioning test plans for a DST-sponsored programme; conducted cross-team engineering reviews.

Automation Lead — Systems & Control  
MARS Rover Team, TIET

Nov 2023 — Feb 2026  
Patiala, Punjab

- Architected complete ROS2 autonomy stack across two successive platforms: designed node topology, defined inter-system interfaces, and owned the control subsystem from specification through field commissioning.
- Implemented multi-loop controllers (position, velocity, torque) with PID tuning under dynamic and non-linear terrain; applied Ziegler–Nichols and optimisation-based methods.
- Redesigned distributed IPC architecture, cutting end-to-end system latency from ~100 ms to ~10 ms—enabling reliable real-time autonomous actuation during field trials.
- Integrated Livox LiDAR, ZED 2i stereo camera, IMU, and custom encoder drivers into a sensor-fused navigation pipeline achieving sub-centimetre localisation accuracy.
- Led software and controls sub-teams; represented the team at international competition reviews and post-mission technical debriefs.

Joint Secretary / Overall Coordinator  
MARS Rover Team, TIET

Jul 2024 — Aug 2025  
Patiala, Punjab

- Scaled engineering team from 22 to 60 members; secured 20L+ in project funding (6× growth); coordinated multi-domain robotics development across controls, perception, and mechanical teams.

KEY PROJECTS

Mars Rover Ganga — IRC 2026  
1st Place, International Rover Challenge 2026

2025 — 2026

- Led systems architecture for full ROS2 autonomy stack: defined subsystem boundaries, data flow, and real-time timing contracts between perception, planning, and control nodes.
- Achieved sub-centimetre navigation accuracy and 75% autonomous mission success across unstructured outdoor terrain; system latency held at ~10 ms throughout field operations.
- Designed hybrid control framework coupling PID position control with perception-driven adaptation; authored interface documentation covering all sensor drivers and control outputs.

## Mars Rover Alaknanda — IRC 2025

2024 — 2025

### 2nd Place, International Rover Challenge 2025

- Built ROS2 navigation system using visual odometry (Intel D435i) and depth-based obstacle avoidance; achieved 10 cm positional accuracy with 65% path-planning success rate.
- Maintained stable closed-loop operation at ~0.63 m/s across multi-terrain field testing within a 100 ms latency budget.

## Lower-Limb Exoskeleton — DST-Sponsored

2025 — Ongoing

### Real-Time Embedded Control, Mentor: Dr. Ashish Singla

- Designed 100 Hz control loop with inner/outer loop architecture; implemented impedance control delivering up to 48 Nm active assistive torque with  $<5^\circ$  steady-state tracking error.
- Integrated biomechanical sensing (Qualisys, AMTI, EMG) for real-time intent estimation; validated system behaviour across user trials before deployment.

## Delta Robot Surgical Platform (DeltaMed)

2025 — Ongoing

### Parallel Manipulator — Kinematics & Master-Slave Control, Mentors: Dr. S. Kansal, Dr. S. Pandey

- Optimised workspace kinematics using PSO, GA, and Bayesian optimisation; designed master-slave control architecture driven by encoder and IMU inputs.
- Validated full kinematic and control model in MATLAB/Simulink prior to hardware realisation (V-model development); presented at IFToMM I4SDG, Italy 2025.

## EDUCATION

### B.E. Electronics, Instrumentation and Control Engineering, Thapar Institute of Engineering and Technology

2022 — 2026

Courses: Data Acquisition System Design, Control System Design, Process Dynamics, Digital & Analog Electronics

## PUBLICATIONS & ACHIEVEMENTS

- **IFTToMM I4SDG, Italy 2025** — Presented: *ROS2 Framework Based Design and Evaluation of an Autonomous Robotic System for Real-Time Grid Failure Assessment*
- **IC2E3 2026** (Accepted) — *DeltaMed: Master-Slave Delta Robot Surgical Platform*
- **IEEE RO-MAN 2026** (Submitted) — *Adaptive PID Switching for Autonomous Rovers*
- **1st Place** — International Rover Challenge 2026 | **2nd Place** — IRC 2025 (BITS Pilani, Goa)
- **Winner** — ERC Robotics Hackathon 2024 (45-day nationwide) | **Best Project** — ELC Engineering Internship 2024
- **INOI 2021 Finalist** — Top 228 nationally; one of 7 selected from Chandigarh (IARCS)

## WORKSHOPS & OUTREACH

- **ANRF Prototype Development Workshop, IIT Ropar (2025)** — Delivered robotics and prototype development session to ~40 top students from IITs, CU, and other institutions.
- **Space Robotics Workshop (2024)** — Introduced robotics and space-systems concepts to high school students.
- **Advances in Robotics (AIR) 2025, IIT Jodhpur** — Attendee and participant.